



energy in electron-volts and the beam current in amperes.

### Types of PBWs

There are two broad categories of PBWs, namely, charged-particle beam (CPB) and neutral-particle beam (NPB) weapons. CPB weapons have a set of technological characteristics that are entirely different from NPB weapons. While characteristics of the former make these suitable for use within the atmosphere, the latter are better suited for use in space.

Both endo-atmospheric (used within atmosphere) and exo-atmospheric (used in space) beam weapons have their own technological hurdles to overcome. A particle beam propagating through atmosphere requires having extremely-high power and precisely-defined beam characteristics.

Technologies required for the

development of a suitable power supply and particle accelerators with sufficient power and appropriately-shaped pulses for endo-atmospheric weapons are very complex and involve high risk.

On the other hand, the greatest challenge in the case of exo-atmospheric beam weapons is in the area of beam control. The PBW should not only be able to produce a high-intensity low-divergence particle beam at the exit of the accelerator, it should also have the necessary beam-control mechanism for aiming and beaming at the target, and the ability to detect pointing errors in the beam for applying correction, if required.

Because of these two different sets of demands, endo- and exo-atmospheric devices represent two different types of weapon systems in appearance and operation. Nevertheless, there are certain fundamen-

tal areas of development that are common to both types of PBWs.

**Charged PBWs.** A CPB consists of electrons accelerated to the required energy level in a particle accelerator using a combination of electric and magnetic fields. To be able to destroy the target, particle energy should be high, and so should be the beam current. As an example, a practical electron beam weapon would need to hit a target 1000km away with a 1000-ampere beam with energy of 1GeV for 0.1 millisecond to destroy it.

Particles in the beam have kinetic energies equal to their rest-mass energies, with the result that these would travel with nearly the speed of light. Particle accelerators researched for high-energy physics have high energies and pulsing rates but low beam currents.

On the other hand, particle accelerators related to fusion research

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